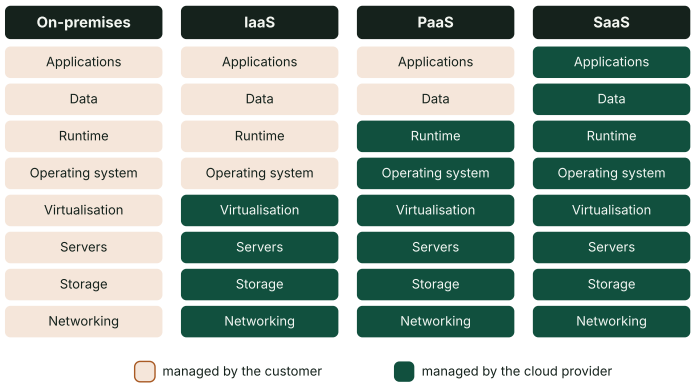
Cloud computing and virtualisation

IS14 FACT FILE

Specification: students should be able to

- Describe cloud service models: infrastructure, platform and software as a service
- Compare public, private and hybrid cloud deployment
- Evaluate the use of cloud computing by organisations
- Explain virtualisation and its benefits

Key words: cloud computing, IaaS, PaaS, SaaS, public cloud, private cloud, hybrid cloud, scalability, virtualisation, hypervisor, virtual machine, data sovereignty



Who manages each layer of the computing stack in on-premises, IaaS, PaaS and SaaS models: the customer or the cloud provider

Service models

Cloud computing provides computing resources over the internet on demand. In infrastructure as a service the provider supplies virtual servers, storage and networking, and the customer installs and manages the operating system and applications. In platform as a service the provider also manages the operating system, runtime and database, so developers only deploy their code. In software as a service the provider runs the whole application and users simply access it, usually through a browser.

Deployment models

A public cloud is owned by a provider and shared by many customers, giving low cost and high scalability. A private cloud is dedicated to one organisation, offering more control and security, which suits regulated sectors such as banking and health, but costs more. A hybrid cloud combines the two: sensitive data and core systems stay private, while public resources handle less sensitive workloads or peaks in demand.

Virtualisation

Virtualisation creates software versions of physical resources. A hypervisor divides one physical server into several isolated virtual machines, each running its own operating system. This consolidates servers, cutting hardware, power and cooling costs, and allows new servers to be created in minutes. Snapshots and live migration support backup and disaster recovery. However, a failure of the host affects all its virtual machines, and there is some performance overhead. Virtualisation underpins cloud computing.

T Key terms from the mark schemes

Cloud computing
The on-demand delivery of computing resources such as servers, storage and software over the internet, usually paid for as used.

Infrastructure as a service (IaaS)
A cloud model in which the provider supplies virtualised hardware such as servers, storage and networking, and the customer manages the operating system and software.

Platform as a service (PaaS)
A cloud model in which the provider supplies the hardware, operating system and development platform so customers only build and manage their own applications.

Software as a service (SaaS)
A cloud model in which complete application software is hosted by the provider and accessed by users over the internet.

Hybrid cloud
A cloud deployment that combines a private cloud with public cloud services, allowing data and applications to move between them.

Virtualisation
The use of software to create virtual versions of hardware, allowing several virtual machines to run on one physical computer.

Hypervisor
Software that creates, runs and manages virtual machines on a physical host computer.

Exam focus

Discussion questions on cloud computing are often extended-response with QWC. Organise your answer around benefits (scalability, cost, access, maintenance, recovery) and drawbacks (connectivity, security, legal responsibility and data location, control, lock-in, ongoing cost), always applied to the organisation in the question. Distinguish service models by who manages what. A balanced judgement such as 'a hybrid approach would suit the trust because…' is needed for the top band.

? Check yourself

1. Describe what is meant by software as a service (SaaS) and give an example. [2]
2. Explain the difference between IaaS and PaaS. [2]
3. Describe a hybrid cloud and give one reason an organisation might choose it. [2]

Answers: 1. SaaS: the complete application is provided over the internet by the provider, who manages the software, updates and infrastructure; the customer manages only their data/settings (1) / example: Microsoft 365, Gmail, Google Classroom (1). 2. IaaS provides virtualised infrastructure – servers, storage and networking – and the customer installs and manages the operating system, runtime and applications, e.g. renting virtual machines (1) / PaaS provides a platform including the OS, runtime and database, so the customer only develops and manages their own applications and data, e.g. deploying a web app to Azure App Service (1). 3. A hybrid cloud combines private cloud (or on-site) infrastructure with public cloud, with data and applications moving between them (1) / reason, any one: keep sensitive data in the private cloud while using the public cloud for less sensitive work or for peak demand (cloud bursting) (1).